CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
CHANGA

B.Tech.(CE/EE/IT) Sem-1 Examination, 2009-10

CE – 101 Fundamentals of Computing & Programming

Date: 16/12/2009, *Wednesday* **Time: 2:00 p.m to 5:00 p.m** **Maximum Marks:70**

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1(a) Write the statements and mention whether the following are True or False. If false also provide the reason. **05**

1. *main* ( ) is a library function declared in stdio.h file.
2. sizeof is an operator as well as keyword in C.
3. float a[10]; This declaration occupies 80 bytes of memory.
4. Two pointers cannot be added.
5. We can create a new file using *fopen*( ) function in C.

Q.1(b) What is the correct output of the following code? **06**

(1)
```
void main ( )
{ int c = 10;
  for ( ;c!=12;)
  printf ("%d\n",c++);
  printf ("%d",c);
}
```

(2)
```
void main ( )
{ int x=3,y=5, z;
  int *m=&x,*n=&y;
  z = *m;
  *m =*n;
  *n = z;
printf("%d\n%d",x,y);}
```

(3)
```
void main ( )
{ int a=3+3/2%2;
  int b=a--;
  int c=b/a;
  printf("%d\n",a);
  printf("%d\n",b);
  printf("%d",c); }
```

Q.2 Write programs in C. **12**

1. Write a program to find the sum of individual digits of a given integer.
2. Write a program to reverse the case of characters of an entered string using the build-in functions of *ctype.h* file. (Don't use ASCII concept.)

OR

Q.2 Write programs in C. **12**

1. Write a program to check whether the entered number is prime number or not.
2. Write a program to enter and sort the array of 10 integers in ascending order.

Q.3 Answer the following questions. **12**

1. Explain entry-controlled and exit-controlled loop with example.
2. What are the advantages of user-defined function?
3. Explain basic structure of a C program.

OR

Q.3 Answer the following questions. **12**

1. Explain else if ladder with example.
2. Explain the different types of operators in C.
3. Explain *Call by value* with example.

P.T.O.

SECTION – II

Q.4 Do as directed.

1. Replace the code using if…else statements. 02

```
que = (q1>q2) ? ((q1>q3)? q1 : q3 ) : q2;
```

2. Rewrite the code using *for* loop statement. 02

```
int m=4,n=4, mn, nm;
nm=6; mn=nm;
   Lab:
   if(nm!=m && nm<=n)
   {  m=n++;
      nm--;
      goto Lab;
   } printf("%d",nm);
```

3. Find out keywords from the following statement. 02

   The merchant told to servant "if you do not switch off the bulb I will break your legs."

4. Write the most appropriate answer. 02

   (i) Any string in C ends with____ symbol which is known as_____ character.

   (ii) 2 occupies ___ bytes and '2' occupies ___ bytes in memory.

5. Classify between valid and invalid variable name in C. 03

   (i) keyword (ii) size-of (iii) ASCII

   (iv) int_var (v) structure.pointer (vi) 2Two

Q.5 Write programs in C. 12

1. Write a program to copy a given string into another string without *string.h* file.
2. A structure has data members: meter and cm. Write a program that enter two variables, add them and assign into third variable. Validate and print the answer. (e.g. 120cm=1 meter and 20 cm)

OR

Q.5 Write programs in C. 12

1. Write a program to evaluate the following series using user-defined function *fact*( ) that returns the factorial of an argument.

   $$\frac{1}{1!}+\frac{2}{2!}+\frac{3}{3!}+...+\frac{n}{n!}$$

2. Write a program to find the maximum from the given array of 10 integers using pointer variable. (Use pointer to enter and to compare elements)

Q.6 Answer the following questions. 12

1. Differentiate between structure and union.
2. Explain strlen( ), strcmp( ), strcpy( ) functions.
3. What is pointer? How is it initialized?
4. Which are the modes of a file? Why the file should be closed in a program?

OR

Q.6 Answer the following questions. 12

1. Differentiate between local and global variable.
2. Explain & and * operator with respect to pointer.
3. Explain fseek( ), ftell( ) functions related to file.
4. Explain unary, binary and ternary operators in C.

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